

Rapid mechanical changes in crab nerve and squid axon during action potentials*

Ichiji TASAKI and Kunihiko IWASA

Laboratory of Neurobiology, National Institute of Mental Health, Bethesda, Maryland 20205 U.S.A.,
and Marine Biological Laboratory, Woods Hole, Massachusetts U.S.A. 02543.

SUMMARY :

1° By using a Fotonic sensor, it was found possible to record rapid mechanical responses of the crab nerve without a signal averager.

2° In squid giant axons, the peak of swelling was shown to coincide fairly accurately with the peak of the action potential.

3° Associated with propagation of an action potential along a squid giant axon, there is a small decrease followed by an increase in the length of the axon.

4° It was emphasized that both mechanical and birefringence responses of the squid axon are diphasic.

5° The origin of the mechanical responses in axons is discussed on the basis of the theories proposed by LOEB, HÖBER and TEORELL.

Key-words : Mechanical changes. Crab nerve. Squid axon. Action potential.

INTRODUCTION

The existence of rapid mechanical movements of the nerve membrane associated with the process of excitation was implicit in the theory proposed by LOEB (1900) and HÖBER (1920). Based on their discovery of marked changes in the physicochemical state of various biocolloids produced by replacement of Ca-ions for Na- or K-ions in the colloids, they proposed that « ion-exchange » processes involving Ca-ions and alkali metal ions underlie excitation phenomena. HÖBER described the resting nerve membrane to be in a Ca-rich, compact state and the excited (or depolarized) nerve membrane to be in a loosened (*i.e.* swollen) state. Much later, based on the results of extensive analyses of his electro-hydraulic model, TEORELL (1962) suggested that the process of action potential production is accompanied by swelling and shrinkage of the nerve membrane resulting from movements of water molecules.

In the electrochemistry of artificial cation-exchanger membranes, the effects of substituting univalent for di-

valent cations on the water-content of the membrane are well understood. As a rule, replacement of univalent for divalent cations increases the water-content and profoundly decreases the membrane resistance. KATCHALSKY and ZWICK (1955) demonstrated profound swelling of a polyelectrolyte gel associated with exchange of univalent ions for divalent ions. It is also well known that the membrane water-content is the most important factor that determines the electric resistance of the membrane (see *e.g.* LAGOS and KITCHENER, 1960; KÖRÖSY and ZEIGERSON, 1968).

Attempts at experimentally demonstrating mechanical changes of the nerve membrane associated with nerve excitation are not new. HILL (1950) and BRYANT and TOBIAS (1955) have shown that prolonged repetitive stimulation of nerve fibers brings about a slow increase in fiber diameter and a decrease in fiber length. HILL *et al.* (1977) reported that there was a « contraction » of the fiber diameter toward the end of an action potential of the crayfish giant fiber. Quite recently we have shown, by the use of three different techniques, that the diameter of nerve fibers increases during the rising phase of the propagated action potential (IWASA *et al.*, 1980; IWASA and TASAKI, 1980).

In this article, we describe several aspects of our recent findings concerning mechanical responses of nerve fibers. We emphasize that the nerve membrane swells during the rising phase of the action potential and shrinks during the falling phase. We show that this diphasicity of the mechanical changes in the nerve membrane is an expression of general physicochemical processes which take place in the membrane when an action potential is generated.

MATERIAL AND METHODS

Both claw nerves of the crab, *Callinectes sapidus*, and giant axons of the squid, *Loligo pealei*, were used in most of the experiments. The artificial seawater was prepared in accordance with the formula recommended by Marine Biological Laboratory, Woods Hole, Mass. and had the following composition : 423 mM NaCl, 9 mM KCl, 9.3 mM CaCl₂, 23 mM MgCl₂, 25 mM MgSO₄ and 7 mM tris-(hydroxymethyl)-

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aminomethane (at pH 8.0). After excision, the nerve preparation was introduced into a plastic chamber provided with two pairs of platinum electrodes (see Fig. 1, top), one pair being used for external stimulation of the nerve and the other pair for recording action potentials extracellularly. The interpolar distance of each of the electrodes was 4–6 mm, and the separation between the two pairs was usually 12 mm. Mechanical changes of the nerve fibers were recorded at a point half-way between the two pairs of the extracellular electrodes. In experiments with squid giant axons, action potentials were recorded intracellularly using a 50 μ m diameter enameled silver-wire electrode with its recording tip placed directly at the site of mechanical recording.

To record mechanical changes of the nerve fibers, both a Fotonic sensor, purchased from Mechanical Technology, Inc., and a piezo-ceramic bender, generously supplied by Gulton Industries, Inc., were employed. The Fotonic sensor is an optical device which consists of two bundles of fiber-optics mixed thoroughly at one (sensing) end: one bundle was used for transmitting light from a quartz-iodine lamp to the sensing end (see the diagram in Fig. 1); the other bundle was utilized for carrying the light from the sensing end to a photo-diode (Pin-10, United Detector Corp.). A light-reflecting target, either a short piece of fine platinum wire or a small piece of gold leaf, was placed on the surface of the nerve. When the distance between the target and the sensing end of the optical device was between 40 and 100 μ m, the intensity of the detected light increased sharply with the distance; based on this distance-dependence of the light intensity, movements of the target associated with production of an action potential was calibrated.

The Gulton bender was used with a small stylus attached to its sensing end. The voltage developed by pressing the stylus was approximately 0.5 mV per mg. The area of the tip of the stylus making contact with the nerve was roughly 10^{-3} cm². To detect variations in the pressure developed by the nerve, the tip was lowered roughly 100 μ m below the point of contact with the surface. A Nicolet signal averager (Model SW71B) was employed to record the pressure variations associated with action potentials.

The experimental setup used for detecting change in the length of the squid giant axon during excitation is illustrated diagrammatically on the right-hand side on Fig. 2, top. An approximately 40 mm long axon was placed vertically in a plastic chamber provided with two partitions near its bottom. These partitions were made of a 3 mm thick plastic plates and the separations between the two partitions was roughly 4 mm. There were small holes in the partitions through which the axon was passed. These holes were sealed with vaseline after introducing the axon. Stimulating shocks were applied across the lower partition and the propagated action potential was recorded across the upper partition. The upper end of the axon was tied to a Gulton piezo-ceramic bender with a fine thread.

Changes in birefringence during action potentials were measured by introducing a squid giant axon into a shallow layer of seawater held between two cover slips. Polaroid sheets (HN series) were placed, one above and another below the axon. The light from a 100 W quartz-iodine lamp was employed for measurements after passing through a heat-cut filter. A Pin-10 photo-diode was used for detecting changes in the light intensity. The polarizing axes of the two Polaroid sheets were kept at 90° to each other, and the longitudinal axis of the axon was at 45° to the polarizing axes of the Polaroid sheets. Most of the experiments were carried out at room temperature (20–22 °C).

RESULTS

1) Mechanical Responses of Crab Nerve Recorded Directly with an Oscilloscope.

Mechanical responses of the crab nerve were found to be large enough to be examined without using a signal

averager. When the output of the photo-diode connected to the Fotonic sensor was amplified and led to an oscilloscope (see Fig. 1, top), the signal associated with production of a full-sized action potential was much larger than the level of the random noise generated by the mechanical recording device. (The amplifier connected to the photo-diode had a flat response between 1 and 1 000 Hz.) The amplitude of the observed mechanical response was estimated by comparing the voltage variation evoked by nerve stimulation with the distance-dependence of the d.c.-output of the photo-diode. In the record furnished in Fig. 1, left, the amplitude of the observed mechanical response was approximately 400 μ V (at the output of the photo-diode stage); since the d.c.-output of the photo-diode stage increased at a rate of 50 mV per μ m, the peak value of the surface displacement of the crab nerve was estimated to be approximately 100 Å.

During the course of development of the present optical technique a large number of crab nerves were used. In the later stage of our experiments, value ranging 50 to 100 Å were obtained in most cases. [The size and the weight of the target were found to affect the results to some extent.]

In Fig. 1, the downward deflection of the optical trace represents a decrease in the light intensity, namely,

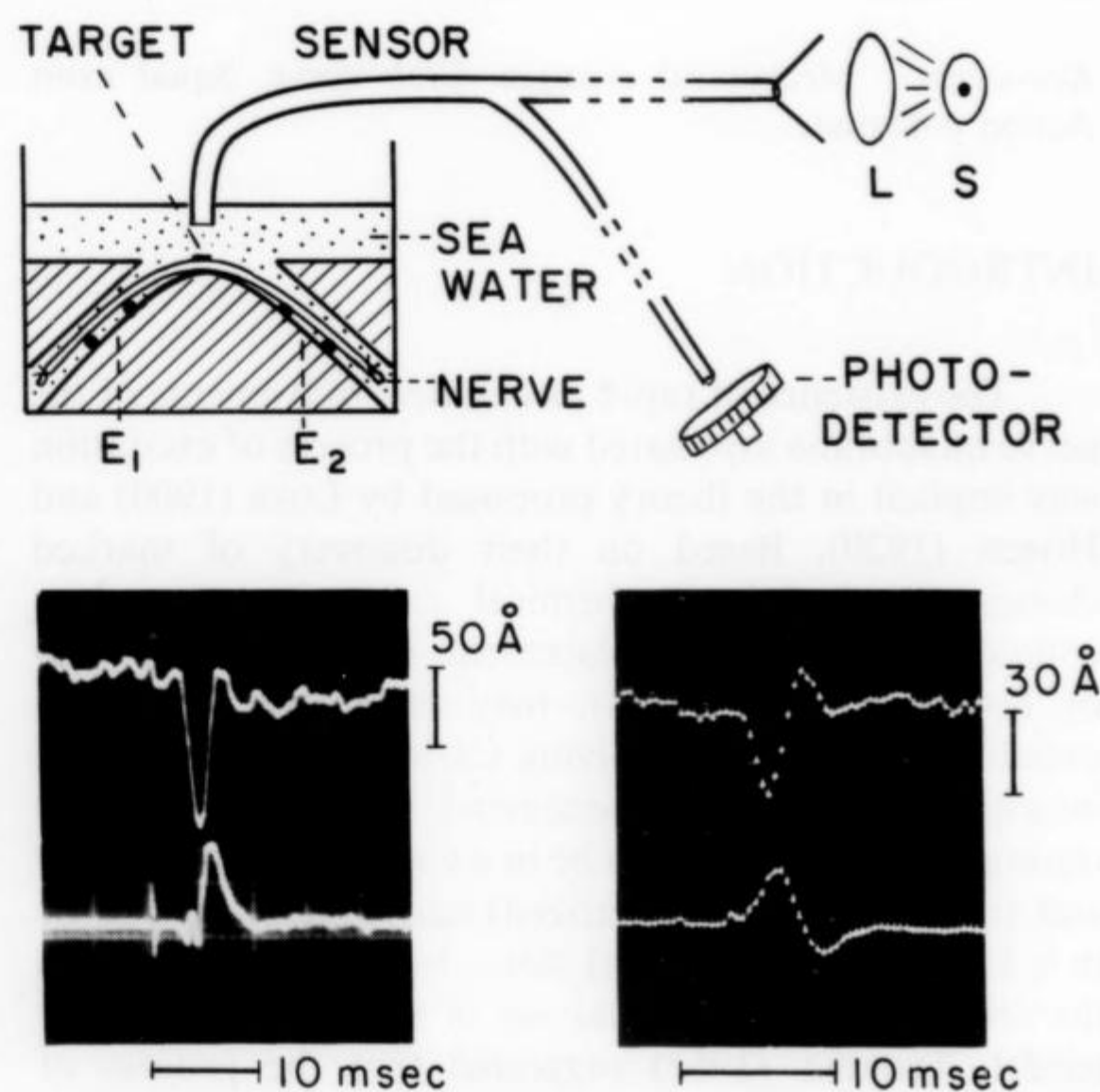


FIG. 1. — *Top*: Diagram illustrating the method of demonstrating swelling of a crab nerve associated with action potential production. Sensor represents two bundles of optical fibers mixed at one end. E_1 and E_2 are stimulating and recording electrodes. *Bottom*: Oscilloscope record indicating a transient decrease in the distance between the light-reflecting target on the nerve and the tip of the sensor (upper trace). Note that the upward movement of the target preceded the onset of the externally recorded action potential (lower trace).

an upward movement of the target associated with production of an action potential (shown by the lower oscillograph trace). This downward deflection was found to start at about the time of arrival of the propagated action potential at the site of mechanical recording. The downward deflection was followed by a small upward deflection which suggests shrinkage of the nerve fibers toward the end of the action potentials. The existence of this phase of shrinkage was demonstrated by the use of signal averager (see the record in Fig. 1, right).

The crab nerve used in these experiments is an assembly of a large number of nerve fibers. An electron-microscopic examination indicated that the fiber diameter varies between 1 and 12 μm , with the peak of numerical predominance at about 2 μm . Since small fibers make a large contribution to the total membrane area, it is probable that small fibers make a large contribution to the mechanical responses. Since large nerve fibers make a relatively large contribution to the extracellularly recorded action potentials, it is very difficult to make an accurate comparison between the time-course of the action potential and the mechanical response. This difficulty can be circumvented by using single giant axons of the squid.

2) Mechanical Responses of Squid Giant Axon.

In order to examine the temporal relationship between the mechanical and electrical responses of nerve fiber, the two type of responses were recorded from the same point of squid giant axon. A long metal wire electrode, which was insulated except at its tip, was introduced longitudinally into the squid axon. Mechanical responses were recorded with a Gulton bender from the portion of the axon directly above the internal recording electrode (see Fig. 2, top left). Since the mechanical responses of the squid giant axon were found to be roughly one order of magnitude smaller than that of the crab nerve trunk, it was necessary in these experiments to

connect a signal averager to the output of the Gulton peizo-ceramic bender.

An example of the mechanical and electrical responses obtained by this method is shown in Fig. 2, left. It is seen in the figure that the rising phase of the mechanical response coincides fairly accurately with the rising phase of the action potential. In all the experiments the peak of the action potential was found to be located within about 50 μsec from the peak of the mechanical response.

In comparing these two different types of responses, it is important to note that the factor which determines the rising phase of the action potential is totally different from that which governs the rapidity of the mechanical response. The rate of rise of the mechanical response is determined primarily by the rate of pressure rise in the axon; the inertia of the recording system is expected to produce some distortion. The rate of rise of the internal potential is determined by the spread of the electronic potential ahead of a propagated action potential. It seems therefore very difficult to make a more accurate comparison between the time-courses of these two types of responses.

Quite recently, we have demonstrated transient changes in the length of the crab nerve at the time when an action potential travels along the nerve (TASAKI and IWASA, 1980). The results of similar observations on squid axons are now described. The technique employed is described under Methods.

The right-hand record in Fig. 2 shows that the tension of a squid giant axon also changes when excited. It can be seen in the figure that the mechanical response recording at the end of an axon is diphasic. The first phase represents in this case an increase and the second phase a decrease in the tension. As expected from the arrangement of the electrodes used for electrical recording, the mechanical response is seen to start roughly at the peak of the extra-

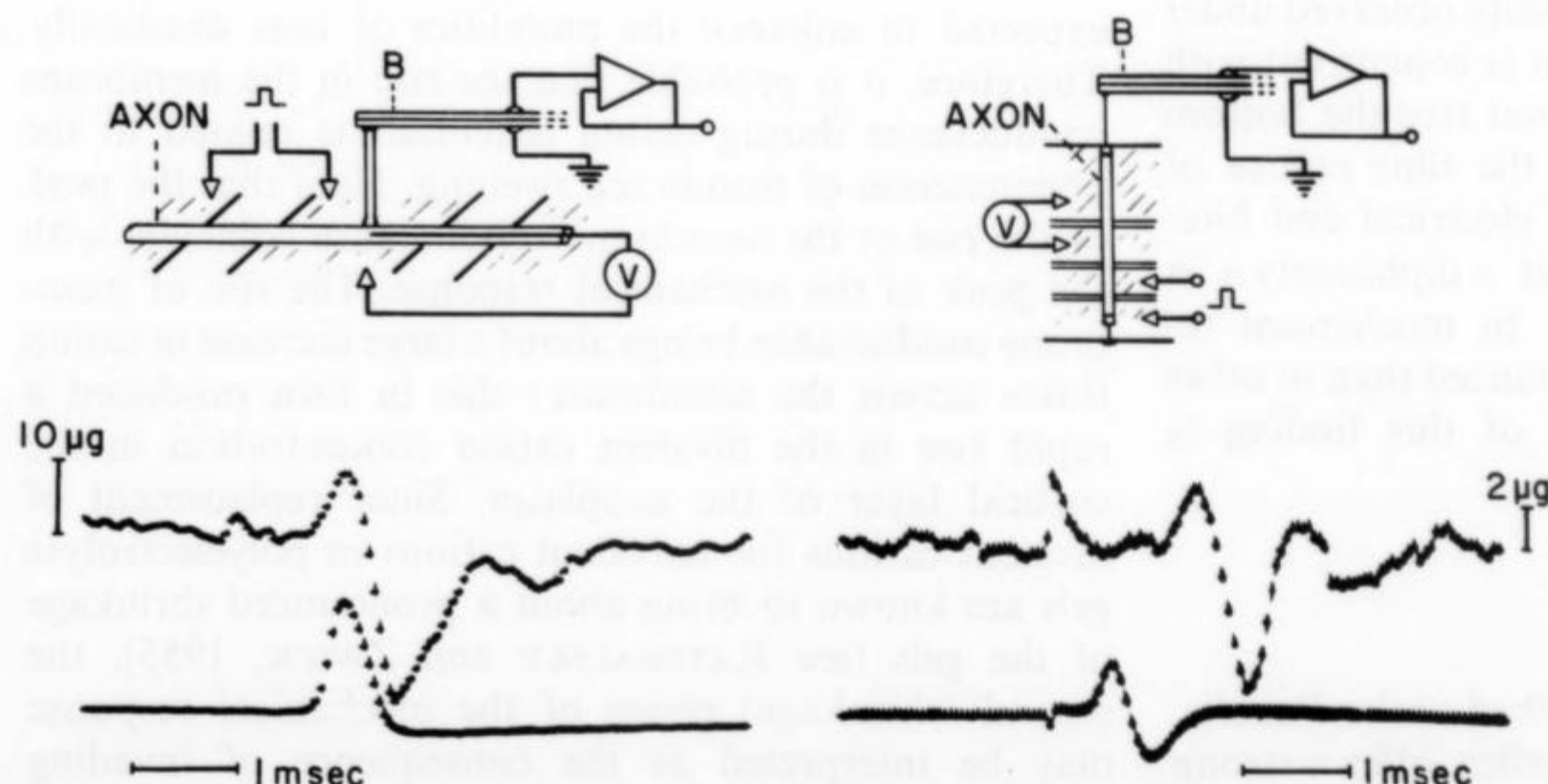


FIG. 2. — *Left top* : Schematic diagram of the set up used for detecting mechanical responses of a squid axon. B represents a Gulton bender. *Left bottom* : Mechanical response (upper trace) and internally recorded action potential (lower trace). *Right top* : Experimental arrangement employed for demonstrating an increase followed by a decrease of the tension of a squid axon associated with action potential production. The portion of the axon in the uppermost pool of seawater was about 15 mm long. *Right bottom* : An example of records obtained by using the arrangement illustrated above.

cellularly recorded action potential. The two phases of the longitudinally recorded mechanical response are considered to reflect the two phases seen in the transversely recorded mechanical response of the axon. In other words, the length of the axon decreases at the time when it swells, and it increases when the diameter decreases.

3) Diphasicity of Mechanical Responses.

It is well known that there is a detectable reduction in the birefringence of a nerve fiber when excited (see COHEN *et al.*, 1971). The time-course of a birefringence change associated with action potential production in the squid axon is diphasic. The peak of a birefringence response coincides fairly accurately with the peak of the intracellularly recorded action potential. The following observation is designed to compare the time-course of birefringence changes with that of mechanical responses.

The upper records in Fig. 3 show an example of mechanical responses obtained by using the Fotonic sensor method. The axon under study was excited simply by reducing the external divalent cation concentration by a factor of 4. (A 3 : 1 mixture of an iostonic NaCl solution and artificial seawater was used in this case; repetitive firing of action potentials induced by this method lasted for a few min.) The mechanical responses were recorded by triggering the signal averager by the action potentials. To facilitate comparison of these mechanical responses with the birefringence responses (shown in the lower part of the figure), the polarity of the mechanical trace was reversed: hence, the downward deflection of the trace in this figure represents an outward displacement of the surface of the axon. It is seen that these responses are highly diphasic, each phase of swelling being followed by a phase of pronounced shrinkage.

Birefringence response (*i.e.* changes in birefringence associated with action potential production) observed in such repetitively firing axons is also diphasic. The downward deflection of this oscillograph beam in the figure represents a decrease in the light intensity observed under cross-polar condition; this deflection is concurrent with the rising phase of the action potential (see the bottom trace in the figure). By comparing the time course of mechanical responses with those of electrical and birefringence responses, it is found that «diphasicity» is common to all of these responses. In mechanical responses, its diphasicity is more pronounced than in other responses. A plausible explanation of this finding is presented in Discussion.

DISCUSSION

The experimental findings described under Results, together with the results published earlier, offer a strong

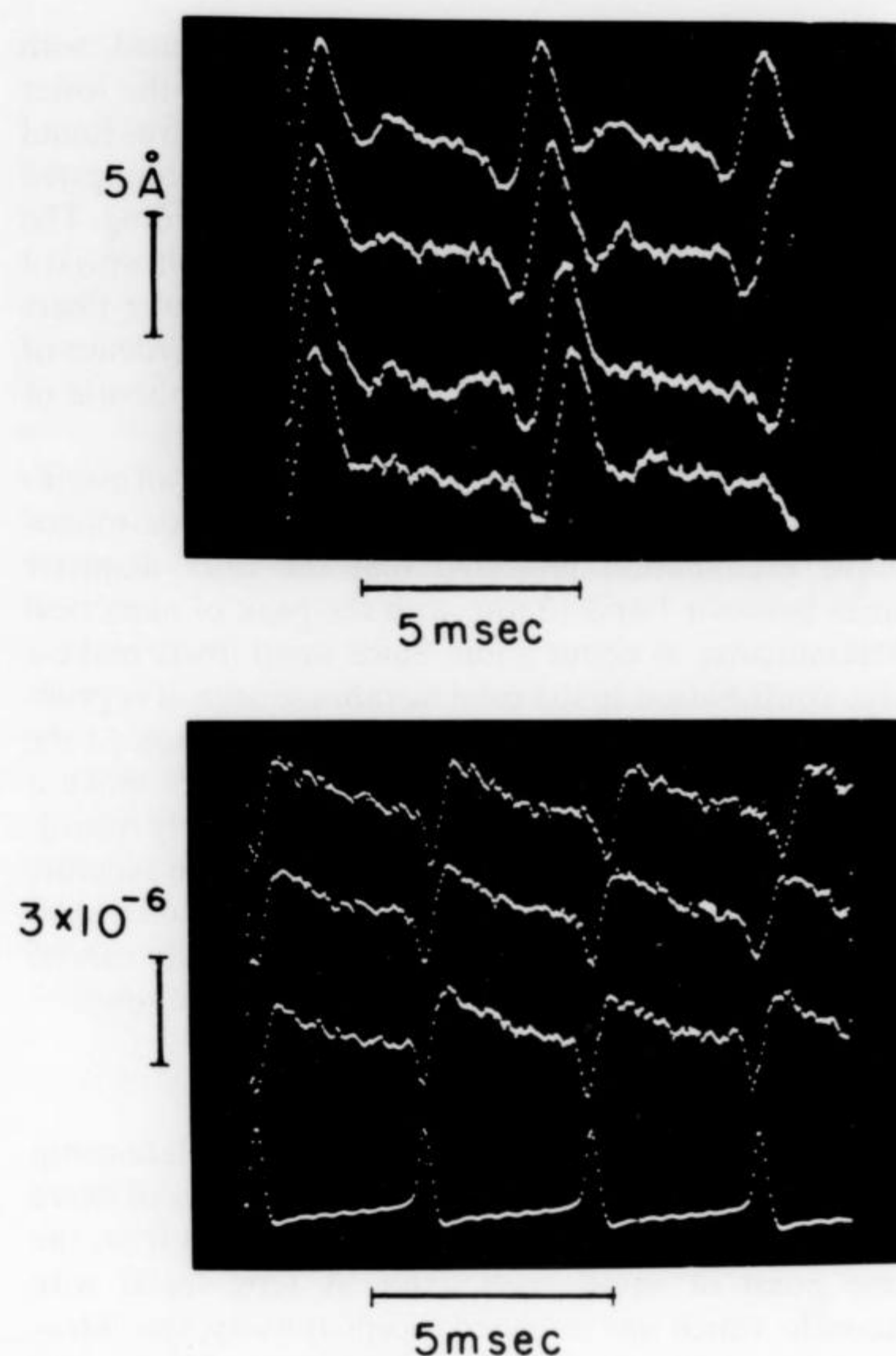


FIG. 3. — *Upper records*: Mechanical responses of a squid giant axon recorded with a Fotonic sensor. The axon under study was firing action potentials repetitively. *Lower records*: Birefringence responses (three upper traces) and action potentials (bottom trace) of a squid axon firing action potentials repetitively.

support to the old concept of LOEB, HÖBER and TEORELL (see Introduction). The swelling of the membrane layer is expected to enhance the mobilities of ions drastically. Therefore, it is probable that the rise in the membrane conductance during action potentials is related to the phenomenon of membrane swelling. Note that the peak of the rise of the membrane conductance coincides with the peak of the mechanical response. The rise of membrane conductance brings about a large increase in cation fluxes across the membrane; this in turn produced a rapid rise in the divalent cation concentration in the cortical layer of the axoplasm. Since replacement of divalent cations for univalent cations in polyelectrolyte gels are known to bring about a pronounced shrinkage of the gels (see KATCHALSKY and ZWICK, 1955), the second (shrinkage) phase of the mechanical response may be interpreted as the consequence of invading

Ca-ions. The endoplasmic reticulum (HENKART *et al.*, 1978) and other calcium-sequestering compartments of the squid giant axon are probably responsible for the termination of the mechanical response.

To understand the mechanism of production of mechanical responses on a molecular basis, it is essential to elucidate the arrangement of various macromolecules in and near the axolemma in greater detail. Therefore, we believe that a further speculation as to the origin of mechanical responses at this time is fruitless.

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